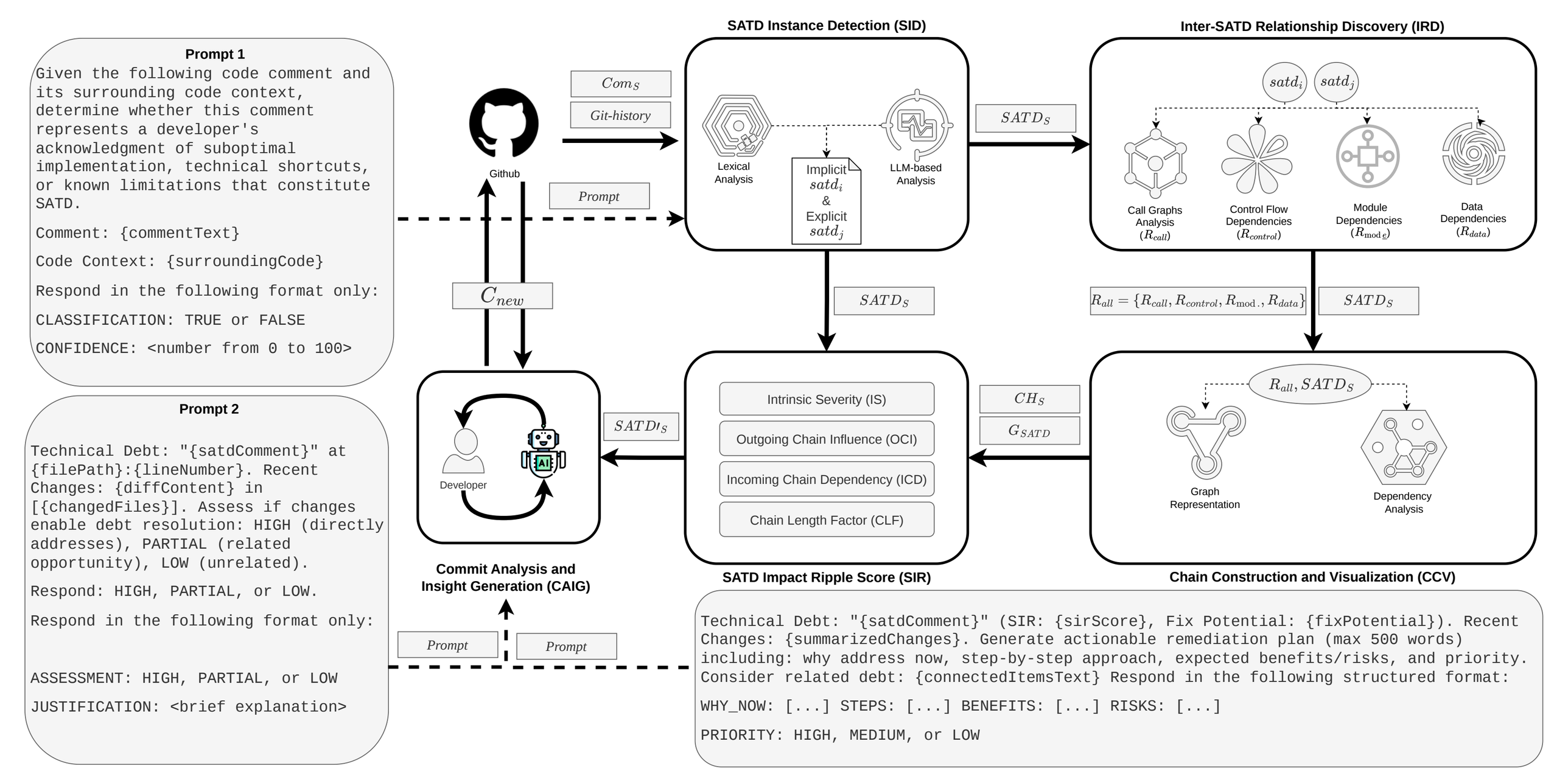

Prompt 1
Given the following code comment and its surrounding code context, determine whether this comment represents a developer's acknowledgment of suboptimal implementation, technical shortcuts, or known limitations that constitute SATD.
Comment: {commentText}
Code Context: {surroundingCode}
Respond in the following format only:
CLASSIFICATION: TRUE or FALSE
CONFIDENCE: <number from 0 to 100>
Prompt 2
Technical Debt: "{satdComment}" at {filePath}:{lineNumber}. Recent Changes: {diffContent} in [{changedFiles}]. Assess if changes enable debt resolution: HIGH (directly addresses), PARTIAL (related opportunity), LOW (unrelated).
Respond: HIGH, PARTIAL, or LOW.
Respond in the following format only:
ASSESSMENT: HIGH, PARTIAL, or LOW
JUSTIFICATION: <brief explanation>
Github
Developer
Prompt
Prompt
Prompt
Prompt
C_{new}
Com_S
Git-history
SATD Instance Detection (SID)
Lexical Analysis
Implicit $satd_i$ & Explicit $satd_j$
LLM-based Analysis
$SATD_S$
Inter-SATD Relationship Discovery (IRD)
$satd_i$
$satd_j$
Call Graphs Analysis (R_{call})
Control Flow Dependencies ($R_{control}$)
Module Dependencies ($R_{\underline{mod}e}$)
Data Dependencies (R_{data})
$SATD_S$
$R_{all} = \{R_{call}, R_{control}, R_{\mathrm{mod}.}, R_{data}\}$
$SATD_S$
$SATD'_S$
Intrinsic Severity (IS)
Outgoing Chain Influence (OCI)
Incoming Chain Dependency (ICD)
Chain Length Factor (CLF)
SATD Impact Ripple Score (SIR)
CH_S
G_{SATD}
$R_{all}, SATD_S$
Graph Representation
Dependency Analysis
Chain Construction and Visualization (CCV)
Commit Analysis and Insight Generation (CAIG)
Technical Debt: "{satdComment}" (SIR: {sirScore}, Fix Potential: {fixPotential}). Recent Changes: {summarizedChanges}. Generate actionable remediation plan (max 500 words) including: why address now, step-by-step approach, expected benefits/risks, and priority. Consider related debt: {connectedItemsText} Respond in the following structured format:
WHY_NOW: [...] STEPS: [...] BENEFITS: [...] RISKS: [...]
PRIORITY: HIGH, MEDIUM, or LOW